

Lab 3 Exercise Solutions

SOCIOL 723, Social Statistics II

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```
library(dplyr)
gss <- readRDS("../Data/gss_earnings.rds")
```

Exercise 1: Bernoulli information

```
set.seed(723)
n <- 200
y <- rbinom(n, size = 1, prob = 0.3)

nll <- function(theta) -sum(dbinom(y, size = 1, prob = theta, log = TRUE))
fit <- optim(par = 0.5, fn = nll, method = "Brent", lower = 0.001,
            upper = 0.999, hessian = TRUE)
th <- fit$par

c(mle = th, ybar = mean(y),
  se_hessian = sqrt(1 / as.numeric(fit$hessian)),
  se_by_hand = sqrt(th * (1 - th) / n))
#>      mle      ybar se_hessian se_by_hand
#>  0.3100  0.3100  0.0327  0.0327
```

The MLE is the sample proportion, and the two standard errors agree to the precision of the numerical derivative. The Hessian is a finite-difference approximation to $-\ell''(\hat{\theta}) = n/[\hat{\theta}(1 - \hat{\theta})]$, the expression derived in lecture, so this is a check on the software, not on the theory.

Exercise 2: Curvature is information

```
set.seed(723)
th_grid <- seq(1.2, 3.0, length.out = 300)
ll_shift <- function(n) {
  y <- rpois(n, 2)
  ll <- sapply(th_grid, function(t) sum(dpois(y, t, log = TRUE)))
```

```

  ll - max(ll)
}
curves <- sapply(c(25, 100, 400, 1600), ll_shift)

matplot(th_grid, curves, type = "l", lty = 1, lwd = 2,
        col = c("grey70", "steelblue", "navy", "firebrick"),
        ylim = c(-40, 0), xlab = expression(theta), ylab = "log-likelihood, peak at 0")
legend("bottomright", legend = paste0("n = ", c(25, 100, 400, 1600)),
      col = c("grey70", "steelblue", "navy", "firebrick"), lwd = 2, bty = "n")

```

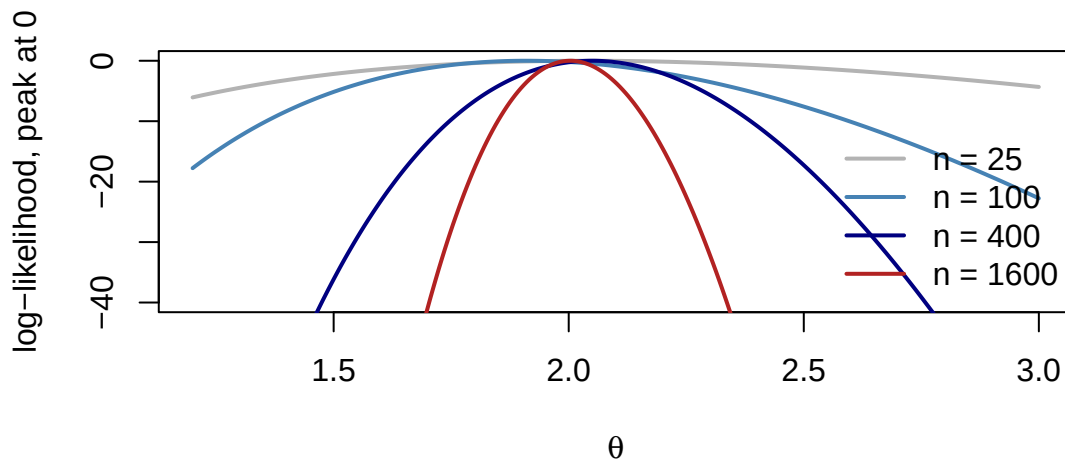


Figure 1: Poisson log-likelihoods for four sample sizes, each shifted to peak at zero.

Each time n quadruples the peak gets visibly sharper: moving the same distance from $\hat{\theta}$ costs four times as much fit. That is $\mathcal{I}_n = n \mathcal{I}_1$ read off the picture: the second derivative is a sum over observations, so it scales with n , and the width of the peak, which is $1/\sqrt{\mathcal{I}_n}$, halves each time.

Exercise 3: Newton–Raphson by hand

```

S <- function(th) 5 / th - 1
H <- function(th) -5 / th^2

th <- 1.5
out <- NULL
for (t in 0:4) {
  step <- -S(th) / H(th)
  out <- rbind(out, c(guess = th, slope = S(th), minus_l2 = -H(th), step = step))
  th <- th + step
}
round(out, 2)
#>      guess slope minus_l2 step

```

```
#> [1,] 1.50 2.33 2.22 1.05
#> [2,] 2.55 0.96 0.77 1.25
#> [3,] 3.80 0.32 0.35 0.91
#> [4,] 4.71 0.06 0.23 0.27
#> [5,] 4.98 0.00 0.20 0.02
```

The table matches the slide. The step is the ratio slope $\div(-\ell'')$. Near the peak $-\ell''$ is at its smallest, so the “learning rate” $1/(-\ell''')$ is at its largest, but the slope is heading to zero faster than the learning rate grows, and their product shrinks; the last step lands on $\hat{\theta} = 5$ almost exactly.

Exercise 4: Not unbiased, but consistent

```
set.seed(723)
sig2_mle <- function(n, reps = 5000) {
  replicate(reps, {
    x <- rnorm(n); y <- 1 + 0.5 * x + rnorm(n)
    e <- residuals(lm(y ~ x))
    sum(e^2) / n
  })
}
s20 <- sig2_mle(20)
s200 <- sig2_mle(200)

rbind(n_20 = c(mean_mle = mean(s20), expected = (20 - 2) / 20),
      n_200 = c(mean_mle = mean(s200), expected = (200 - 2) / 200))
#>      mean_mle expected
#> n_20      0.8986      0.90
#> n_200     0.9883      0.99
```

At $n = 20$ the MLE of σ^2 averages about 0.9, not 1: the $(n - 2)/n$ from lecture. At $n = 200$ the shortfall is one percent. The bias is real and it disappears as n grows, which is exactly what maximum likelihood promises: consistency, not unbiasedness.

Exercise 5: Where the z -test fails

```
n <- 20; th0 <- 0.1
ll <- function(theta, s) ifelse(s == 0, 0, s * log(theta)) + (n - s) * log(1 - theta)
rulers <- function(s) {
  th <- s / n
  se <- sqrt(th * (1 - th) / n)
  c(theta_hat = th, se = se, z = (th - th0) / se, W = ((th - th0) / se)^2,
```

```

    LR = 2 * (ll(th, s) - ll(th0, s))
}
round(rbind(zero_successes = rulers(0), one_success = rulers(1)), 3)
#>           theta_hat    se      z      W      LR
#> zero_successes      0.00 0.000   -Inf    Inf  4.214
#> one_success       0.05 0.049  -1.026  1.053  0.668

```

With zero successes the standard error is zero and the Wald ratio is undefined: the test cannot even be formed. The LR statistic exists, $LR = 4.21$, and rejects $\theta_0 = 0.1$. With one success both can be computed, and both accept, but at different values, $W = 1.05$ against $LR = 0.67$, because the log-likelihood at $\hat{\theta} = 0.05$ is nowhere near a parabola: it drops steeply toward zero and gently toward larger θ . The Wald test extrapolates the width at the peak; the LR test measures the actual loss of fit between $\hat{\theta}$ and θ_0 .

Which to trust: lecture's exact table for $n = 20$ says neither is at its nominal 5% here. Zero successes has probability $0.9^{20} = 12\%$ under the null, so any test that rejects it (LR does) has size at least 12%, and any test that cannot score it (Wald) is not a test at all. With two expected successes there is no parabola to read, and the honest answer is an exact binomial test. What survives is the ranking: the Wald ruler is the one that breaks, in both directions, because it reads only the peak; from $n = 50$ onward LR sits nearest 5% and is the one to believe when they disagree.